

Inventory Analytics

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Lynn Khant Minn

Supply Chain Analyst

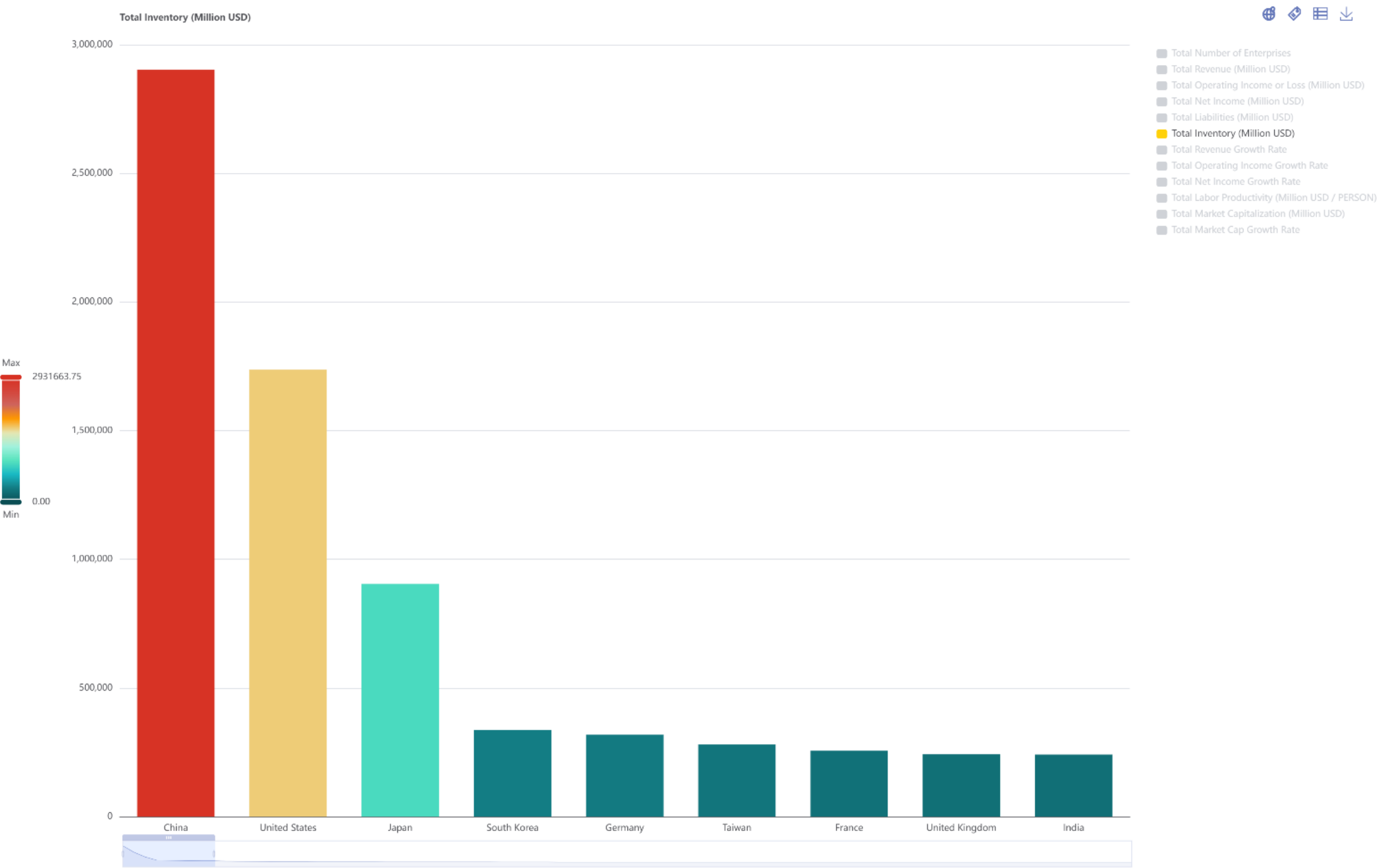
Berlin, Gemany

Objectives

- To identify the industry-country combinations in which inventory is important by **inventory turns** and **inventory days**
- How inventory may impact on the financial performance
- Diagnose the inventory problems for PUMA and Adidas

**To identify the industry-country combinations in which
inventory is important by inventory turns and inventory days**

Enterprise Distribution



The inventory level of China stands first in the world, followed by US, South Korea, Germany, Taiwan , etc.

Enterprise Distribution



In Germany, Bayern state is the highest inventory level.

Enterprise Distribution



In Thailand, Bangkok significantly stands at the top of inventory level.

Inventory / Total Assests

**Although inventory is highly critical for some industries,
some industries do not require inventory**

Benchmarking by KPI - Inventory / Total Assets , Level-1 Industry, Germany



Benchmark by KPI - Inventory / Total Assets

Industry	90th Percentile	75th Percentile	Median	25th Percentile	10th Percentile
Energy-Germany			0.11*		
Materials-Germany	0.35	0.19	0.17	0.13	0.06
Industrials-Germany	0.34	0.25	0.12	0.05	0.02
Consumer Discretionary-Germany	0.32	0.25	0.17	0.08	0.01
Consumer Staples-Germany	0.32	0.29	0.15	0.12	0.08
Health Care-Germany	0.25	0.16	0.09	0.02	0
Financials-Germany	0.05 #	0.02 #	0 #	0 #	0 #
Information Technology-Germany	0.28	0.22	0.03	0	0
Telecommunication Services-Germany		0.02*	0.01*	0.01*	
Utilities-Germany		0.05	0.03	0.01	
Real Estate-Germany	0.59	0.09	0.02	0	0

* The sample size is smaller than 5.

Less than 20% of the sample has value.

Benchmarking by KPI - Inventory / Total Assets Level-1 Industry, Thailand

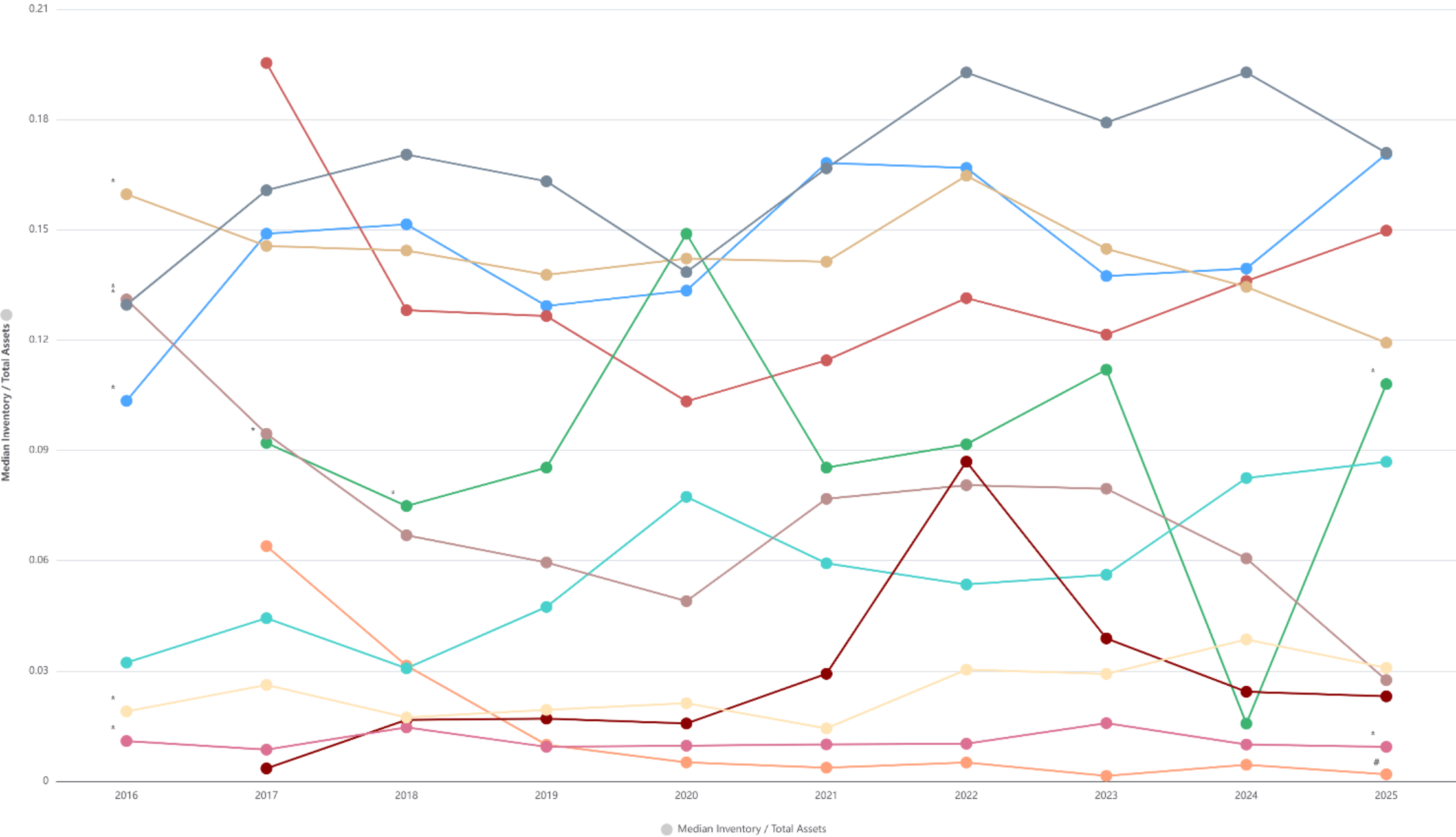


Benchmark by KPI - Inventory / Total Assets

Industry	90th Percentile	75th Percentile	Median	25th Percentile	10th Percentile
Energy-Thailand	0.15	0.08	0.04	0.01	0
Materials-Thailand	0.35	0.22	0.14	0.08	0.05
Industrials-Thailand	0.29	0.18	0.07	0.02	0
Consumer Discretionary-Thailand	0.37	0.22	0.08	0.02	0
Consumer Staples-Thailand	0.31	0.22	0.14	0.09	0.04
Health Care-Thailand	0.27	0.15	0.02	0.01	0.01
Financials-Thailand	0.16 #	0.01 #	0 #	0 #	0 #
Information Technology-Thailand	0.28	0.14	0.06	0.02	0
Telecommunication Services-Thailand		0.01	0.01	0	
Utilities-Thailand	0.08	0.03	0.01	0.01	0
Real Estate-Thailand	0.78	0.66	0.47	0.15	0.01

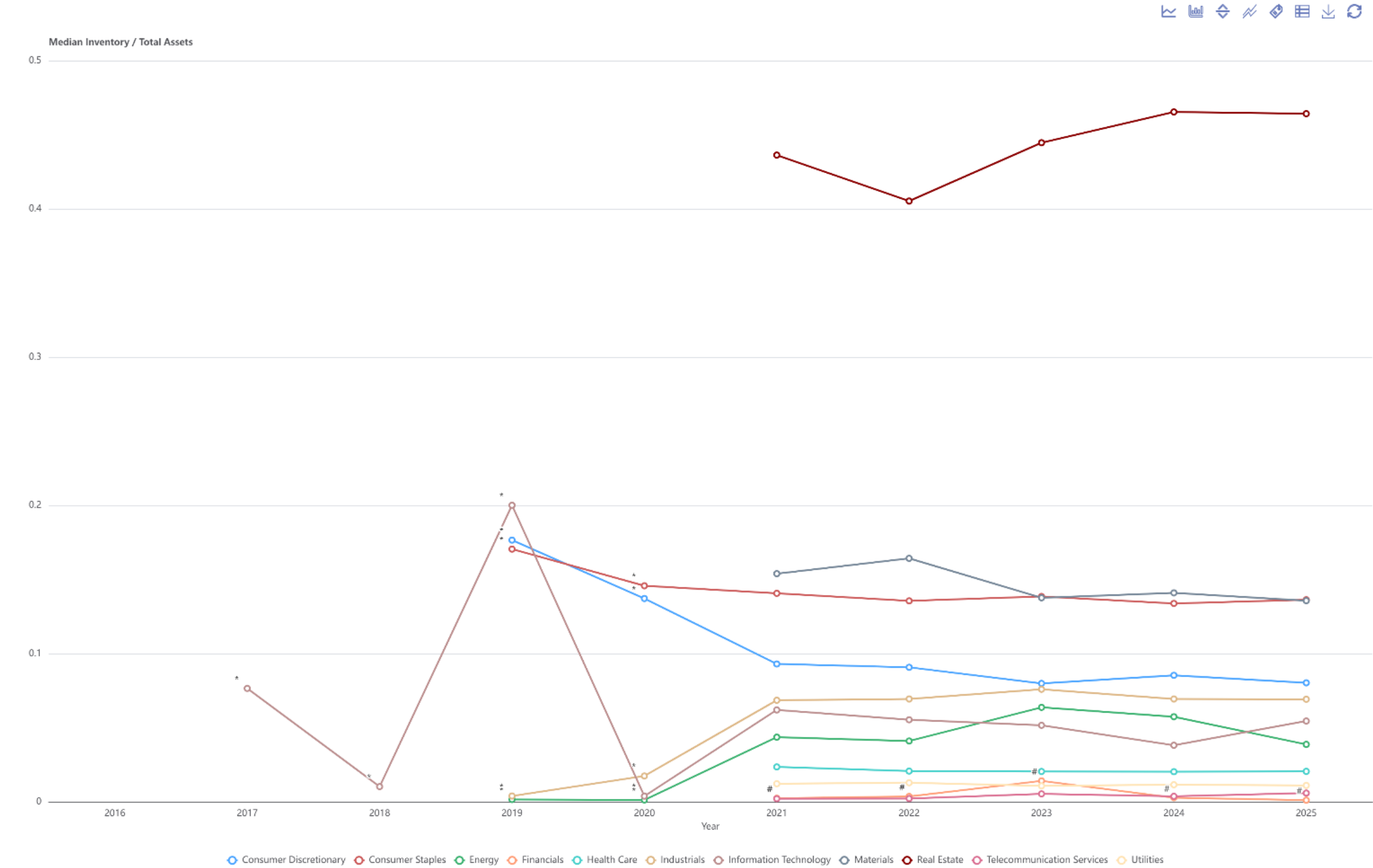
Less than 20% of the sample has value.

Industrial Trend - Inventory / Total Assests (Germany)



* These KPI and Year combination(s) have a sample size smaller than 5. # Less than 20% of the sample has value.

Industrial Trend - Inventory / Total Assests (Thailand)



* These KPI and Year combination(s) have a sample size smaller than 5. # Less than 20% of the sample has value.

Benchmarking by KPI - Inventory Days (Germany)



Benchmark by KPI - Inventory Days (Day)

Industry	90th Percentile	75th Percentile	Median	25th Percentile	10th Percentile
Energy-Germany			37.94*		
Materials-Germany	260.62	150.43	117.87	78.49	38.97
Industrials-Germany	297.26	142.18	84.37	43.49	5.14
Consumer Discretionary-Germany	253.80	123.57	98.87	57.92	11.66
Consumer Staples-Germany	229.64	166.11	138.51	72.25	28.66
Health Care-Germany	820.56	191.24	104.69	32.34	9.46
Financials-Germany		94.69 #	12.71 #	2.59 #	
Information Technology-Germany	260.78	185.09	38.13	0.80	0.29
Telecommunication Services-Germany		21.84*	18.16*	10.38*	
Utilities-Germany		48.06	27.47	9.61	
Real Estate-Germany	11,431.59	1,207.15	203.85	3.25	1.23

* The sample size is smaller than 5.

Less than 20% of the sample has value.

Benchmarking by KPI - Inventory (Thailand)

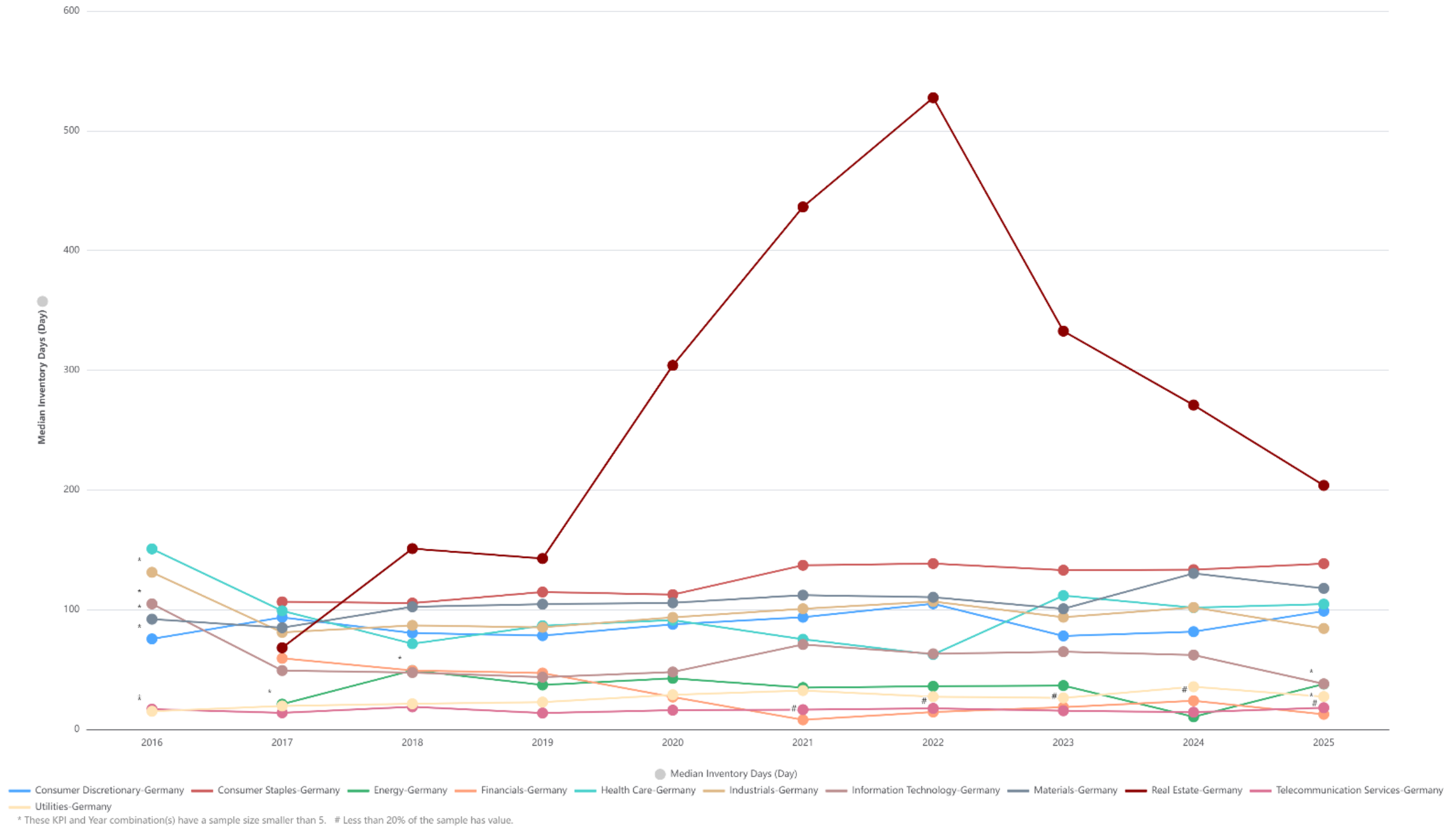


Benchmark by KPI - Inventory (Million USD)

Industry	90th Percentile	75th Percentile	Median	25th Percentile	10th Percentile
Energy-Thailand	1,026.71	361.05	9.42	1.25	0.36
Materials-Thailand	126.84	55.27	17.23	4.57	1.82
Industrials-Thailand	44	14.63	5.04	1.27	0.19
Consumer Discretionary-Thailand	114.99	27.56	6.65	1.59	0.08
Consumer Staples-Thailand	124.38	41.61	15.70	5.83	2.35
Health Care-Thailand	28.92	11.90	2.35	1.10	0.35
Financials-Thailand	16.62 #	5.43 #	0.20 #	0 #	0 #
Information Technology-Thailand	116.89	19.26	4.69	0.69	0.07
Telecommunication Services-Thailand		59.21	1.77	0.48	
Utilities-Thailand	120.71	35.60	3.40	1.43	0.44
Real Estate-Thailand	1,100.26	433.86	123.28	24.40	1.45

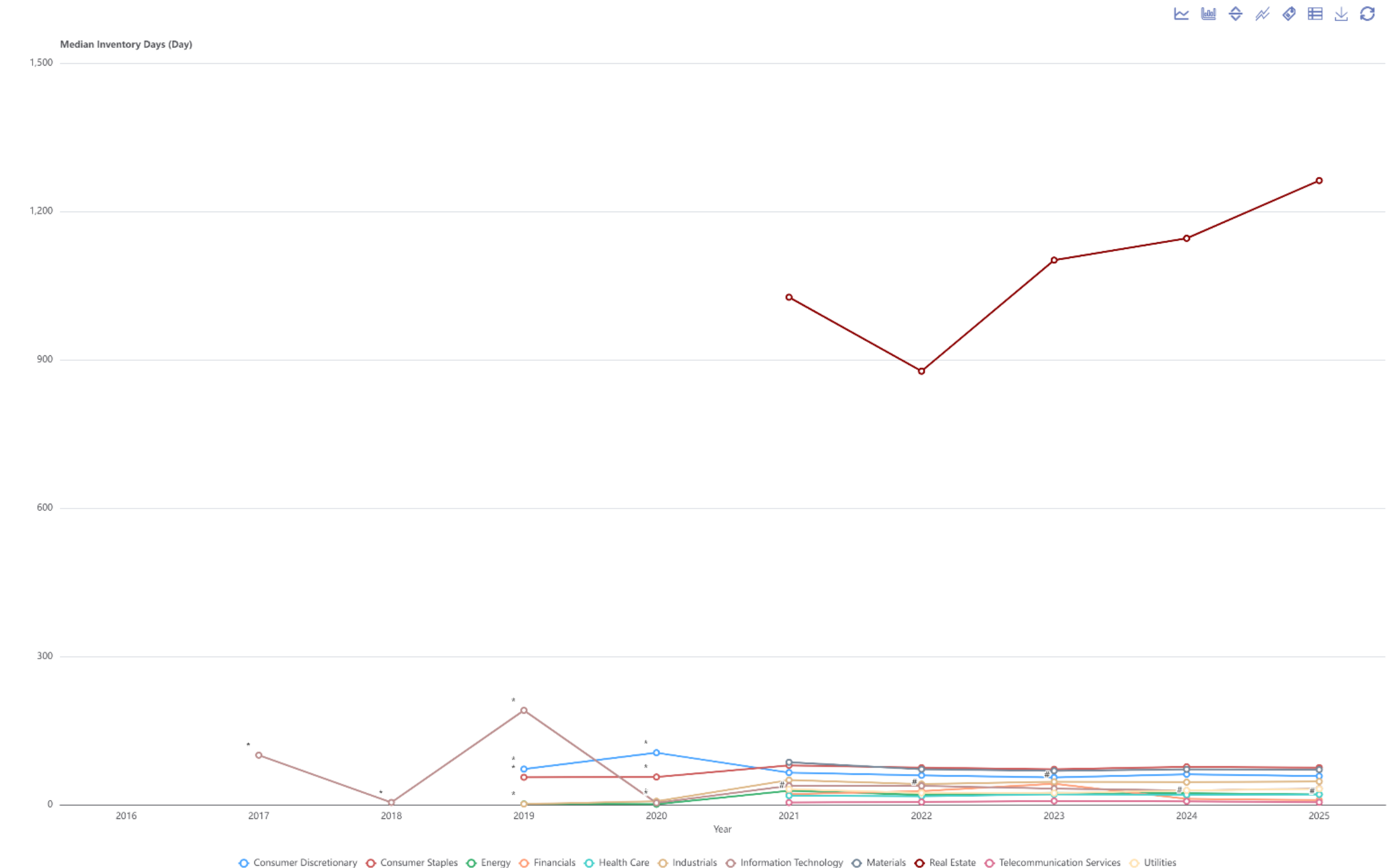
Less than 20% of the sample has value.

Inventory Problems Persist Over Time - Germany



In Germany, Real-Estate inventory days were getting higher since 2017 until 2022, and started falling again in 2023. Other highest inventory day industries are consumer staples, materials, etc.

Inventory Problems Persist Over Time - Thailand



In Thailand, Real-Estate industry has the most obvious inventory days followed by consumer staples and materials. Surprisingly, IT inventory days were the second highest in 2019.

Which country/industry is inventory critical ...?

In Germany-Materials(Median Inventory Days-119)and
Consumer-Dicretionary(Median Inventory Days-99) industries
have the highest median 17%of inventory over total assets in each sector.
Note:Real-Estate inustry has the highest median inventory days 204.

In Thailand-Real-Estate (Median Inventory Days-123) industries have the highest 47% of
inventory over total assets.



How Inventory may impact on financial performance

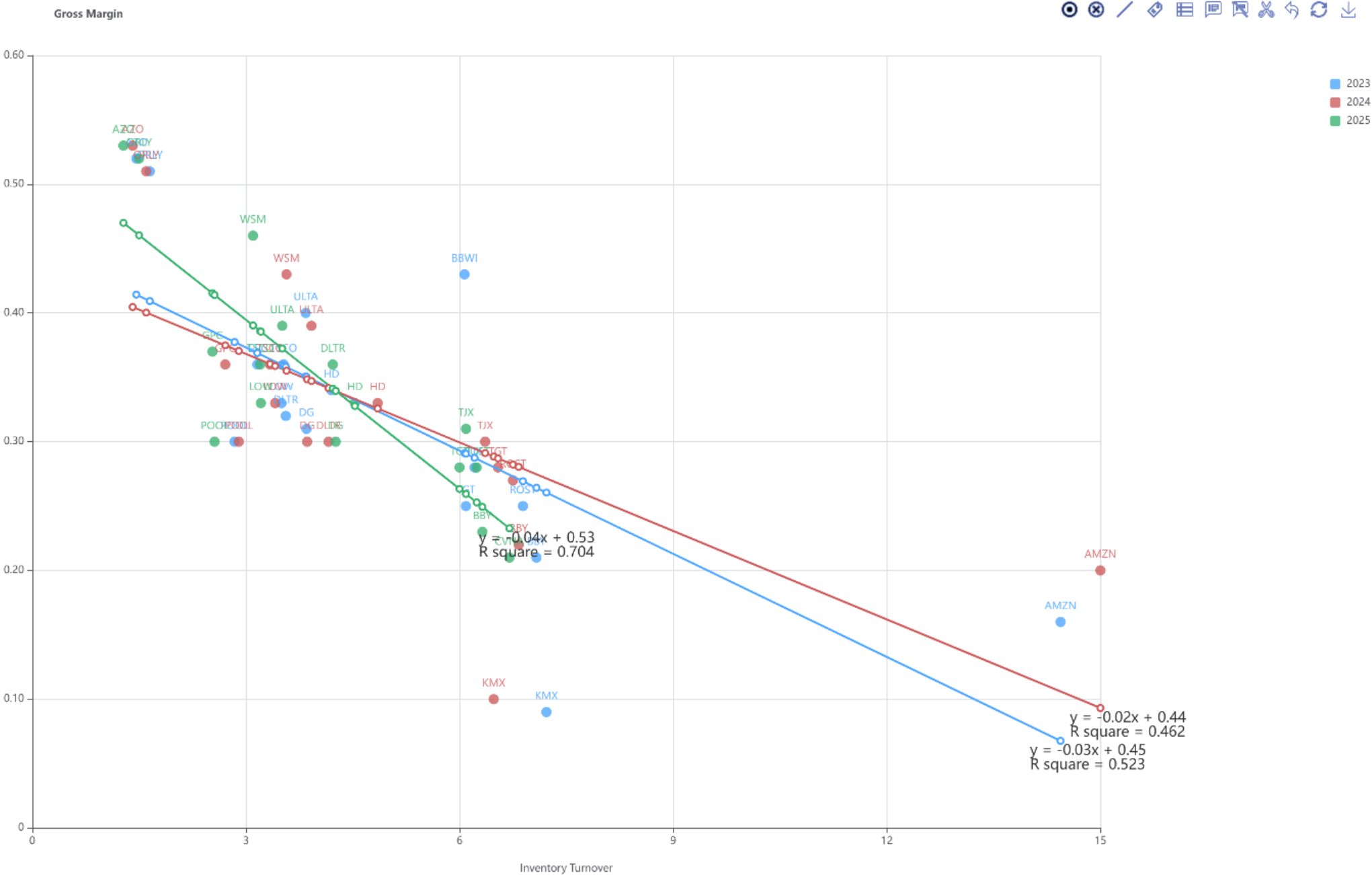
Inventory vs Financials

Value Driver Analysis



In the US, for retailing sector, the inventory turnover improves while SG&A cost over total revenue decrease.

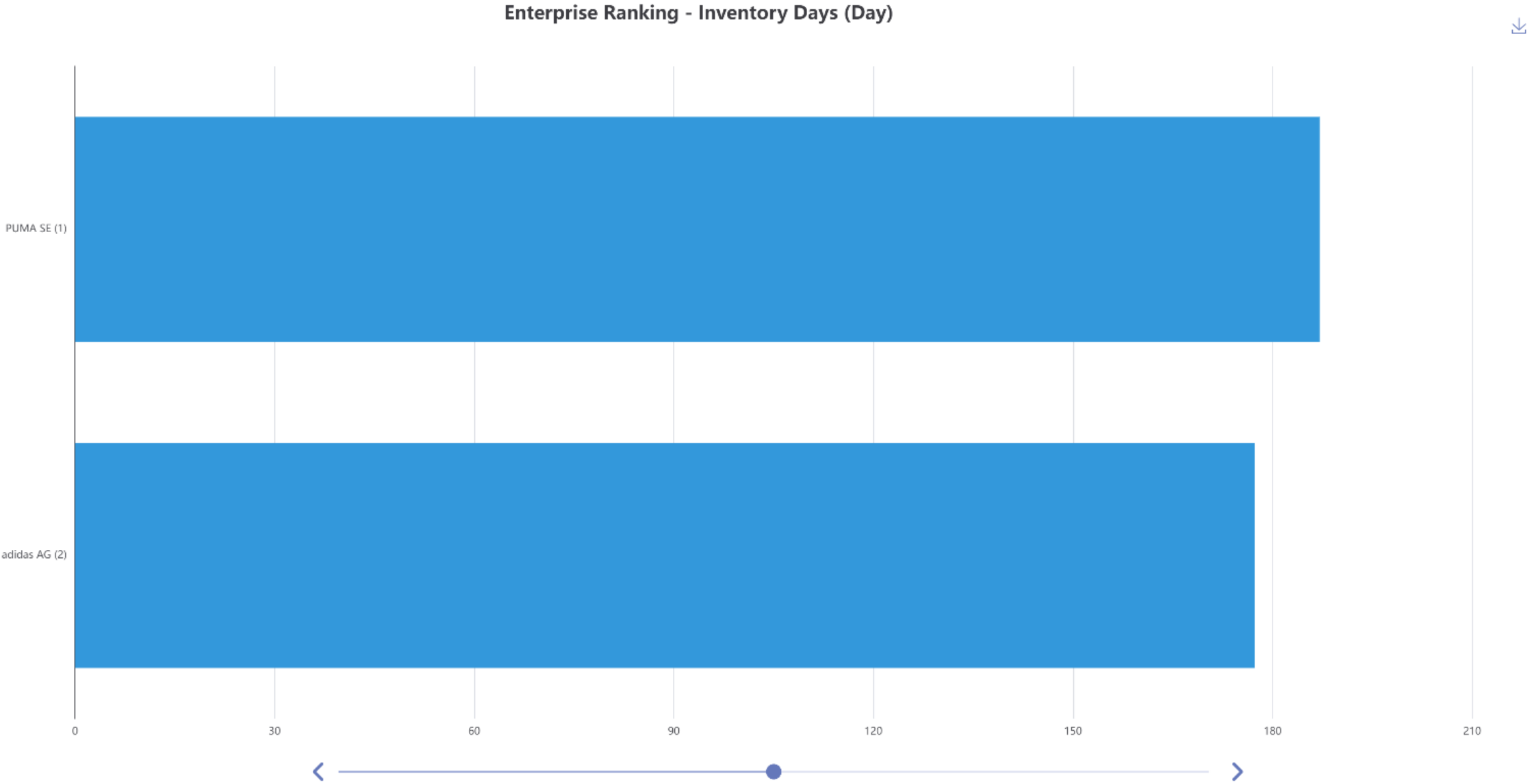
Value Driver Analysis



Lower gross margin means lower pricing, lower-price products are sold faster
meanwhile the inventory turnover rate improved

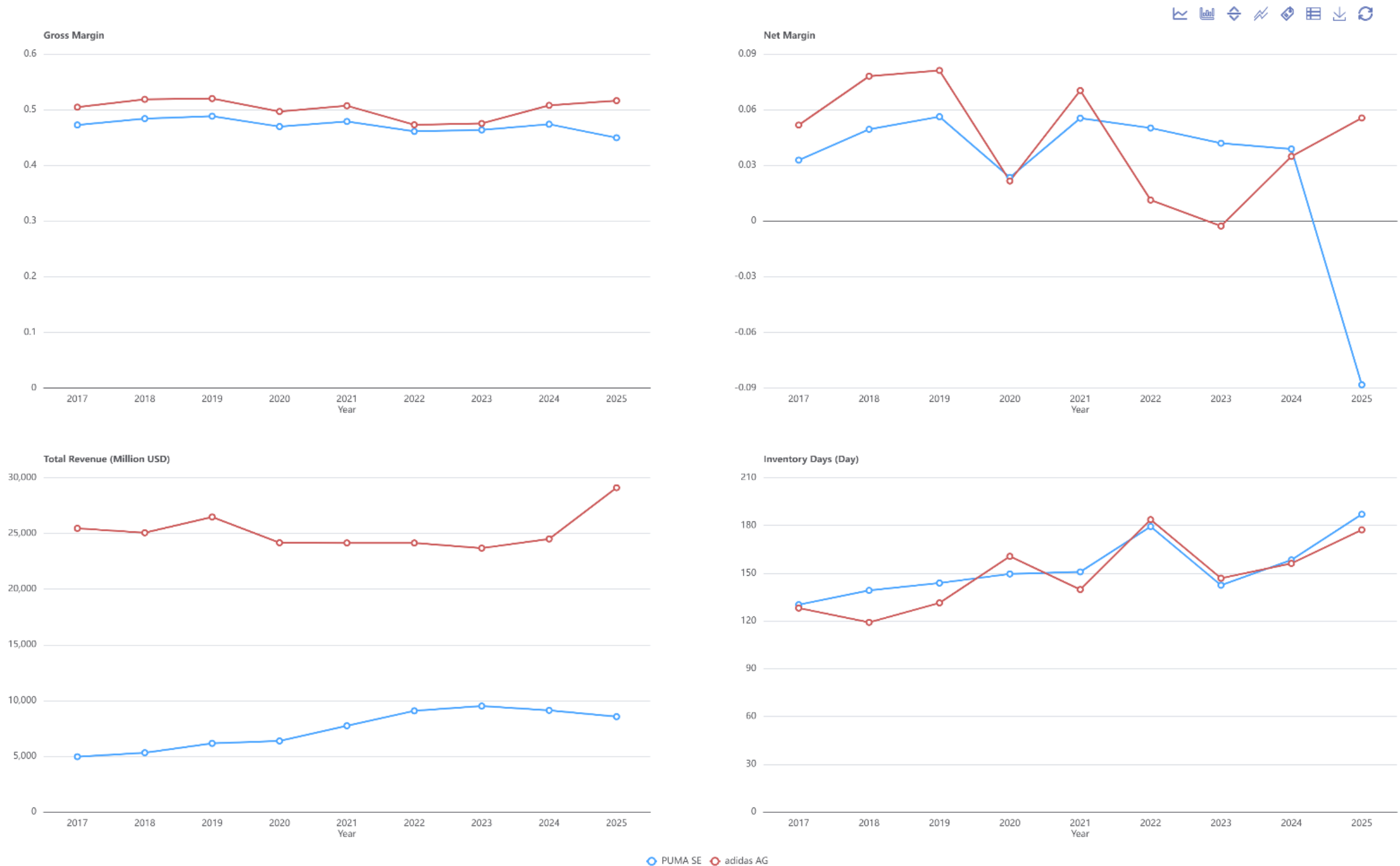
Diagnosing the inventory problems for PUMA and Adidas

Enterprise Ranking - Inventory Days



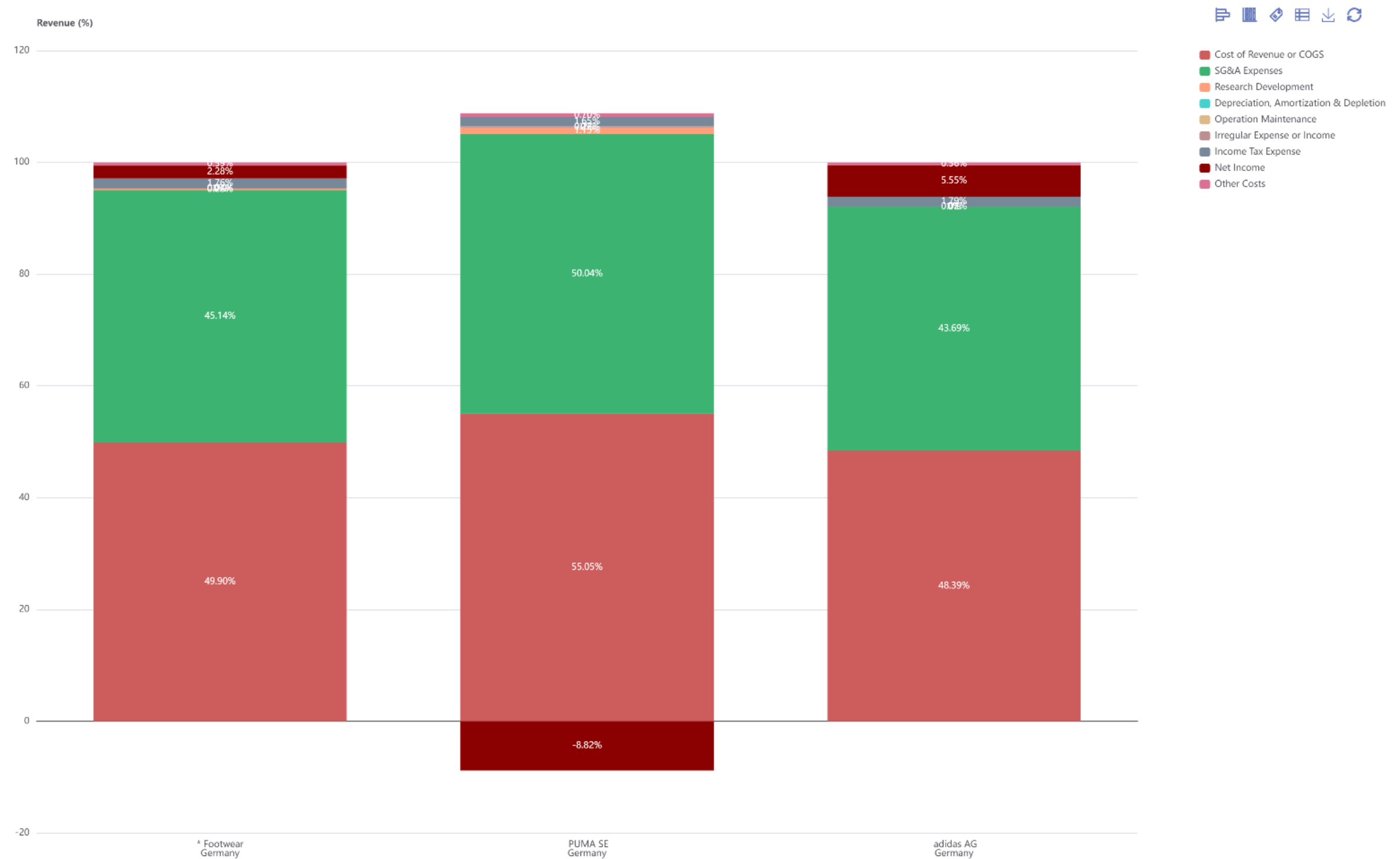
Ranking apparel retailers in Germany by Inventory Days.

Enterprise Trend



As we can see that though the inventory level of PUMA footwear and Addidas footwear is overlapping each other, PUMA has lower total revenue, gross margin and net margin than ADIDAS.

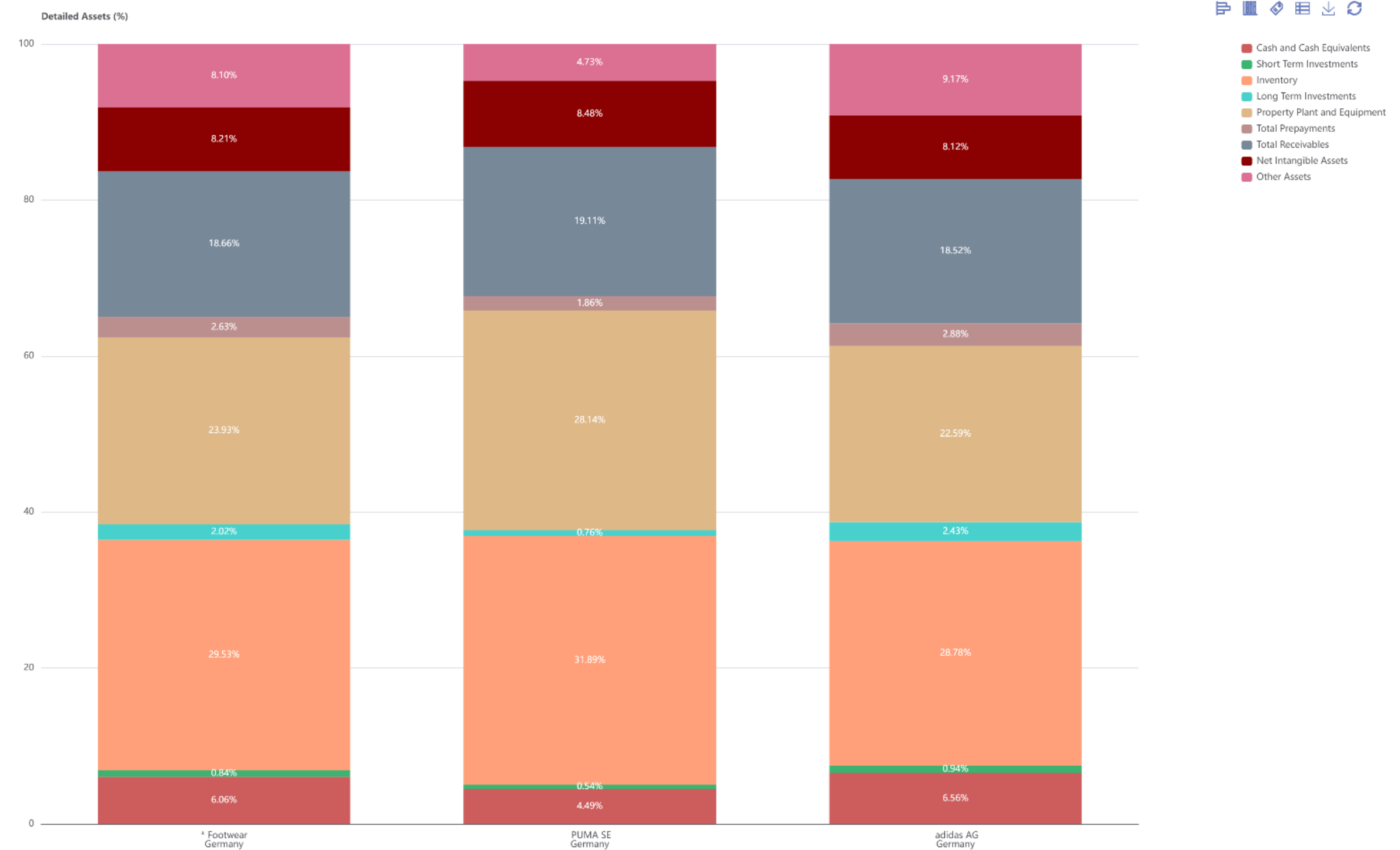
Enterprise Breakdown - Revenue (2025)



* These industry and country combination(s) have a sample size smaller than 5.

According to revenue breakdown, we can see that PUMA even lost profits. Besides, SG&A costs and COGS are higher than Adidas. What contributed to the higher cost of PUMA ?

Enterprise Breakdown - Detailed Assets (2025)



* These industry and country combination(s) have a sample size smaller than 5.

Detailed assets breakdown gives the answer, the inventory of PUMA is higher than Adidas and average footweae industry in Germany. Financial analysis theory suggests that faster inventory reduce SG&A costs. So, inventory is likely to contribute the higher costs of PUMA.

The Comparison of PUMA vs Adidas

PUMA

High profit

Low SG&A cost

Low inventory & cash cycle

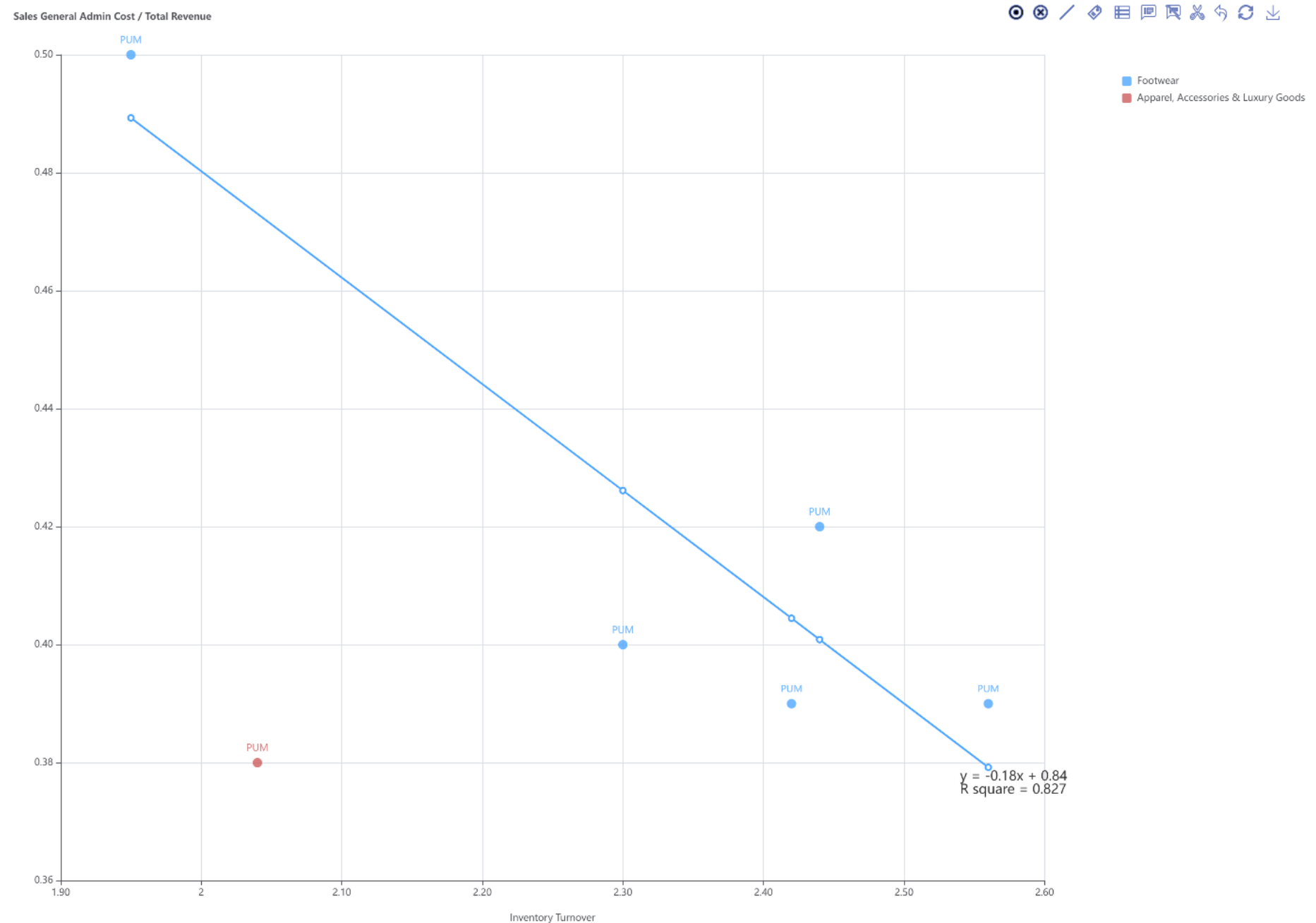
Adidas

Low profit (Even Loss in 2025)

High SG&A cost

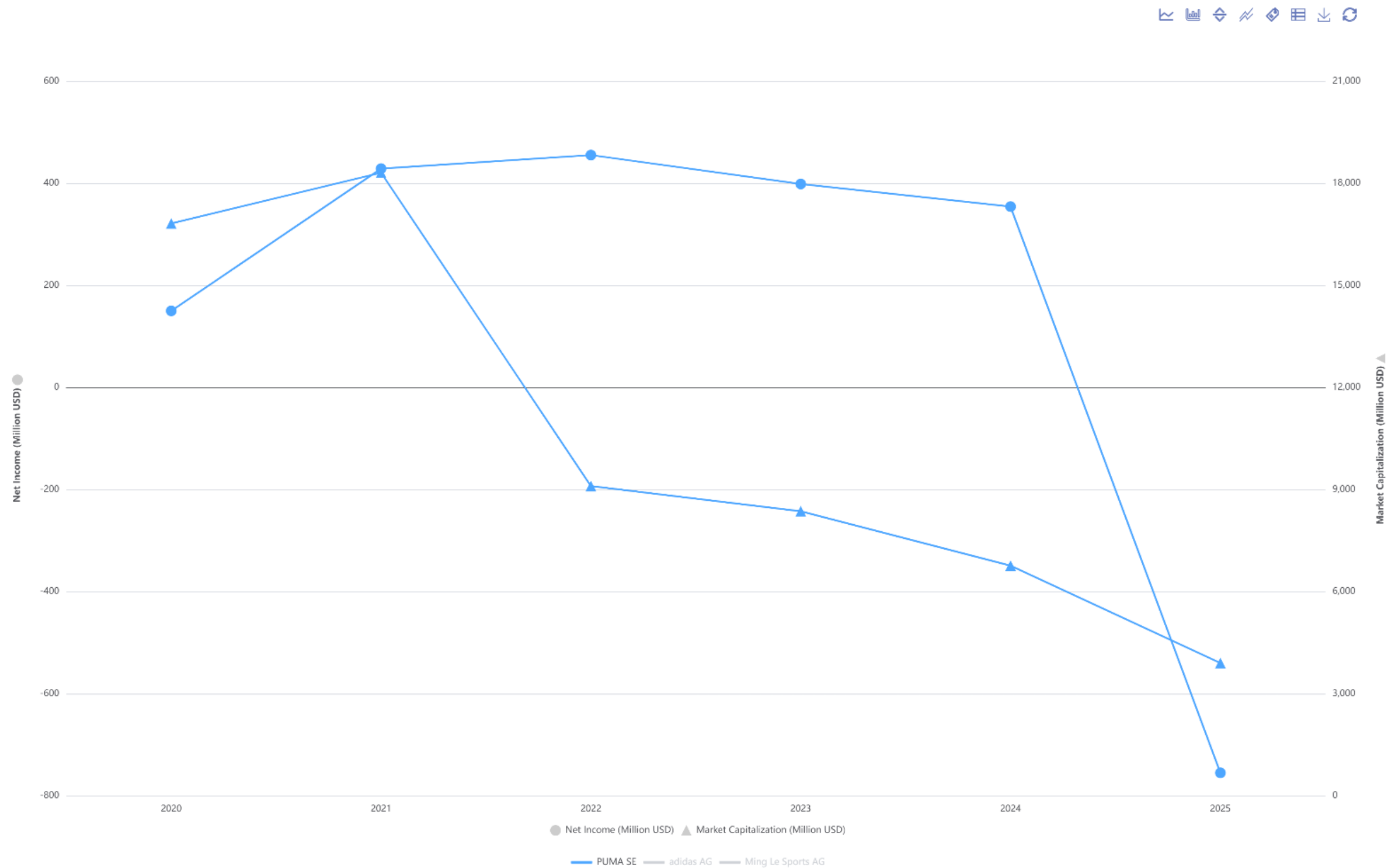
High inventory & cash cycle

Value Driver Analysis



Using the latest 5 years' data, we can see a fair strong negative relationship between inventory turnover and SG&A cost. The data proves that for this footwear company, inventory turnover is a massive value driver. When they manage their supply chain efficiently and move products off the shelves faster (higher x), they spend significantly less on warehousing, administration, and overhead (lower \$y\$). By pushing that inventory turnover higher, they are directly improving their Operating Margin.

Enterprise Trend



As we can see that PUMA's market capitalization goes along with net income. If net income could be improved by inventory management, then market capitalization will also increase.

Recap

- * Inventory is relevant and critical for some nations, industries and segments along the supply chain. However, it is not that much important in others.
- * Inventory issues are often persistent over time.
- * Inventory can increase the market value of a company.
- * Benchmarking and data analysis can diagnose inventory problems.